GRIDZILLA

A grid signal for home batteries, cars and heat pumps.

Mathew Stafurik · Peter Mikeška · Matthew · Tedi

Grid-Serving Storage · Energy Bridge Vienna 2026

Europe's largest battery is in people's basements.

The grid can use none of it.

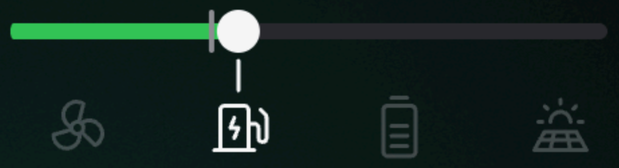
Slovakia



0 signals

from the grid operator
to home batteries

Germany



4.2 kW

the operator may dim
one new charger to this

Ukraine



3 GWh

of home batteries, idle
through the blackouts

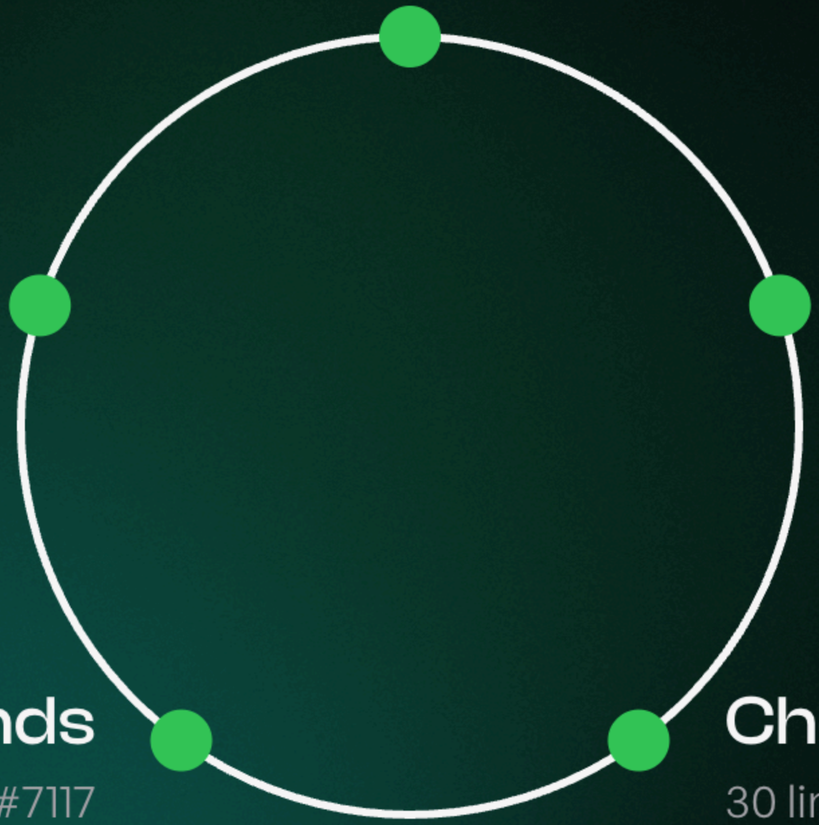
The storage exists. The signal does not.



Battery first. Car second. Hot water third.

The operator asks. The house decides. The owner can always say no.

The loop, every 15 minutes.



Gathers

72 transformers, the weather, the plan, every reply

Thinks

The AI writes the message in 1.5 s

Checks

30 lines of plain rules, not AI

Signs and sends

The platform signs message #7117

Houses reply

271 homes answer, all signed

Six words. Signed.

SUPPORT

Evening peak, or the grid short of power

“Take at most 3 kW from the street.”

ABSORB

Noon sun flooding the street

“Send back at most 2 kW.”

PREPARE

A warning: storm, planned outage, attack

“Be 90 % full by 17:00.”

STRETCH

The street is dark, power back at a known time

“Run on your own, send nothing back, make it last.”

RECOVER

The transformer is replaced

“Refill slowly, not all at once.”

RELEASE

The event is over

“You are on your own again.”

Plus a 24-hour envelope per street, renewed every two hours. Every message Ed25519-signed with a time window, ramps randomised so a street never steps in sync, the owner's limits always win, and everything reverts when the heartbeat stops.

The monster eats peaks.

Every street, every evening.



68%

with Gridzilla, tonight's plan down from 88 % without it

The next 24 hours for Vienna: 2,168 homes, cars and heat pumps, renewed every two hours from the real weather. Live.

Saved tonight

5.7 MW
off the city's evening peak
25.8 MW without the plan, 20.1 MW with it

57 kW
kept off the grid right now
925 kW without Gridzilla, 868 kW with it

2 of 2
streets kept inside their limit
that would pass it tonight without the plan

10,419 kWh
car charging moved into the night
1,184 cars full by morning, 7,064 kWh from home batteries

What we did

0
signals signed today
since 00:00 grid time

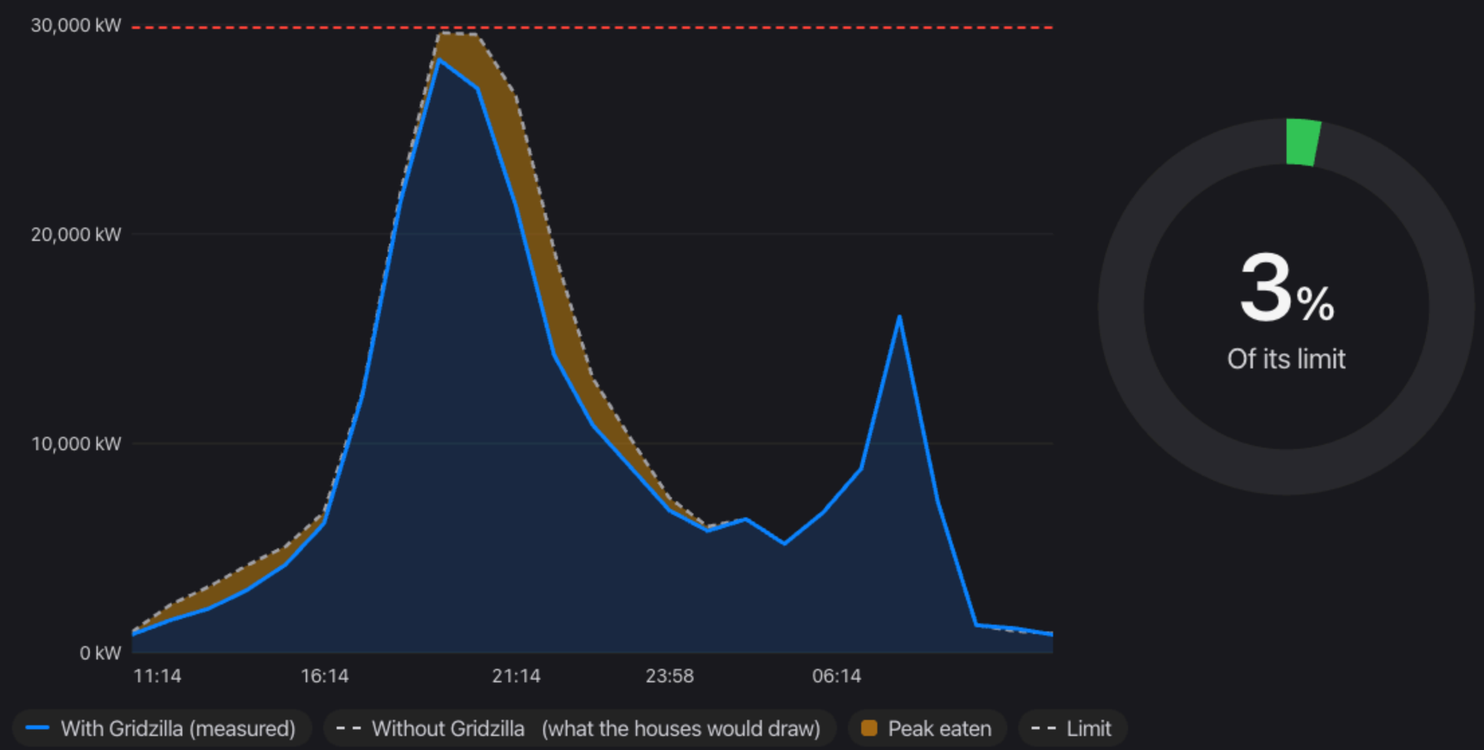
0
houses following a signal now
no refusals today

0
AI decisions today
0 sent, 0 stopped by the guard, diet plan renewed 1x

Vienna · all 72 transformers limit 30 MVA

No transformer in Vienna is forecast past 85 % of its limit. The houses run on their own.

VIENNA | Wien Sued | Wien Nord | Wien West | Wien Ost | Kendlerstrasse | Wien Mitte | Bisamberg | Schwarzenbergplatz



What the AI did

Next review in **off**

1 What the AI saw
86 %
Forecast on Simmering at 23:58

2 What it chose
Use your battery
on Simmering
Simmering is forecast at 86 % around 00:00 as heat pumps and EV chargers come on.

3 Rule check
Allowed
Within the rules: time window, depth, how often, how much power back

4 Signed message
#7117
Use your battery on Simmering, 24 homes
24 applied

Blatnica · real plant · live

● live · updated 14 s ago · Cerbo GX + Vitocal via Gridzilla agents

+0.05 kW
grid import

39 %
battery SoC · 15 kWh · owner floor 20 %

0.0 kW
EV charger · 11 kW station · idle

0.0 kW
heat pump · tank 50.6 °C → 40 °C · room → 21 °C · floor flow 32 °C



owner mode · last ack reverted (window_end) · signature ✓

What just happened

All events

10:18 Diet plan renewed for 24 feeders · 2,168 houses on the plan · a new 24-hour plan per street
04:48 Use your battery on Simmering 28 ended: its time window ran out
03:12 Use your battery on feeder Donaustadt 4 ended: its time window ran out

Our family home in Blatnica

Victron battery	Home at the meter
39%	0.1kW
Car charger	**Viessmann heat pump**
0.0kW	0.0kW

Live, on the same signed message path

Signed
by the platform, one message

Checked
by the plant, signature first

Followed
by the battery and heat pump

Real end to end. The planning runs on the fleet.
Paid by the meter: a flat yearly fee, crisis prices only when the meter proves the limit was kept.

A simulated Vienna

Two thousand houses, the same cars with and without Gridzilla, realistic charging.



72 street transformers in a simulated Vienna, live

72 inside the limit · 0 near it · 0 over

−36%
one street
Atzgersdorf, 511 to 325 kW

−5MW
Vienna
two thousand houses, 31.2 to 26.2 MW

0kW
a street
where no car charged

Measured on the simulated fleet, 24 Oct 2026, with the heat pumps shedding floor heating as well

It eats peaks for breakfast.

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